CCC-C-442
SEPTEMBER 30, 1958
SUPERSEDING
Fed. Spec. CCC-D-730
July 30, 1943
(See 6.5)

FEDERAL SPECIFICATION

CLOTH, COTTON, DUCK, BLEACHED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. — This specification covers bleached duck cotton cloth used in the manufacture of clothing items (see 6.1).

1.2 Classification.—The bleached duck cotton cloth shall be of the following types as specified (see 6.2).

Type I. —5 ounces per square yard.
Type II. —7 ounces per square yard.
Type III.—8.2 ounces per square yard.
Type IV.—8.5 ounces per square yard.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

CCC-T-191—Textile Test Methods.
PPP-P-51 — Packaging, Packing and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.
Fed. Std. No. 102—Preservation, Packaging and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes and Appendix thereto.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Standard sample. — Unless otherwise specified (see 6.2), the cloth shall be equal to the standard sample in color only (see 6.3).

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3.2 Preproduction sample approval.—Unless otherwise specified (see 4.2.2 and 6.2), before production is commenced the contractor shall submit for approval to the contracting officer a two-yard sample of the cloth he proposes to furnish.

3.3 Material.

3.3.1 *Cotton.*—The cotton shall be of suitable length of staple and grade to meet the requirements of this specification.

3.3.2 *Yarn.*—The cotton used in the manufacture of the yarn shall be thoroughly cleaned, well carded, drawn, evenly spun and twisted into the ply as shown in table I.

3.4 Color. — The color shall be bleached white to match an approved shade standard (see 6.3). The use of fluorescent optical bleaches to complement the color of the bleach is permitted.

3.4.1 *Matching.*—The color shall match the approved shade of the standard sample (see 3.1) under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard shade under incandescent lamplight at 2800° Kelvin.

3.5 Physical requirements.—The finished cloth shall conform to the requirements shown in table I.

3.5.1 *Width.*—The width shall be as specified (see 6.2) and shall be the minimum acceptable width exclusive of the selvages.

3.5.2 *Weave.*—The weave for type I and III cloths shall be plain, one up, one down. The weave for type II and IV cloths shall be an oxford weave with two ends weaving as one and alternating in plain weave order with each pick.

3.6 Finish. — The cloth shall be desized, singed, bleached and mercerized. Total residual size, finishing or nonfibrous material shall not exceed 2 percent when tested as specified in 4.2.4.

3.7 Shrinkage. — The cloth shall be preshrunk and shall not shrink more than 1 percent in either the direction of the warp or of the filling when tested as specified in 4.2.4. The preshrinking process used shall not be identified on the cloth, the ticket or on the package.

3.8 Length. — Unless otherwise specified (see 6.2), the cloth shall be in one continuous piece of not less than 50 yards.

TABLE I.—*Physical requirements*

Type	Weight per square yard (minimum)	Yarn ply		Yarns per inch (minimum)		Breaking strength in pounds (minimum)	
		Warp	Filling	Warp	Filling	Warp	Filling
	Ounces						
I	5.0	2	2	77	44	130	75
II	7.0	1	2	88	28	110	60
III	8.2	2	2	58	48	135	100
IV	8.5	1	4	112	32	120	120

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3.9 Tickets.—Each bolt or roll shall have a ticket attached to the selvage at the inner end of the cloth with not finer than 5-ply cotton string doubled to not less than 8 inches long. The ticket shall be made of cardboard not less than 0.020 inch in thickness and the color shall be white or light in intensity to permit easy reading of printed or penciled markings. The finish on the cardboard shall be suitable for pencil markings. The ticket shall have a reinforced eyelet for attaching the tying cord and shall be legibly printed with water insoluble ink with the following information and in addition, the ticket shall have sufficient blank space for the inspector's stamp.

Nomenclature
Type_____Color_____Width_____
Stock number
Contractor's name
Contract number and date
Specification number and date
Name of contracting office
Contractor's piece number
Gross yards
Net yards
Inspector

3.10 Workmanship.—The cloth shall conform to the quality and grade of product established by this specification, and the occurrence of defects shall not exceed the levels set by the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Nonmilitary. — Nonmilitary agencies may use sampling, inspection, and test procedures outlined under 4.2 when desirable. When procedures outlined under 4.2 are not deemed suitable, the procedures specified under nonmilitary paragraphs shall be followed.

4.1.1 *Sampling.*

4.1.1.1 *Inspection lot.*—Unless otherwise specified in the invitation for bids, all cloth of one type submitted at one time shall be considered a lot for purposes of inspection and tests.

4.1.1.2 *Sampling for inspection.*—A random sample of cloth shall be taken from each lot in accordance with table II.

TABLE II.—*Sampling for inspection*

Number of yards in lot	Number of yards in inspection sample [1]	Acceptance number, total (defective)	Rejection number, total (defective)	Acceptance number, major (defective)	Rejection number, major (defective)
300 and under	35	7	8	2	3
301 to 500	50	9	10	3	4
501 to 800	75	13	14	4	5
801 to 1,300	110	18	19	6	7
1,301 to 3,200	150	24	25	8	9
3,201 to 8,000	225	34	35	11	12
8,001 to 22,000	300	44	45	14	15
22,001 to 110,000	450	62	63	20	21
110,001 to 550,000	750	98	99	31	32
550,001 and over	1500	184	185	56	57

[1] Yards in the inspection sample shall be divided equally among sample rolls of cloth (table IV) and that number of yards shall be unrolled from each sample roll.

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4.1.1.3 *Sampling for tests.*—Random samples for tests shall be taken from each inspection lot in accordance with table III.

TABLE III.—*Sampling for tests*

Number of rolls or bolts in lot	Number of rolls or bolts in sample[1]	Acceptance number (defective)	Rejection number (defective)
8 and under	2	0	1
9 to 15	3	0	1
16 to 40	4	0	1
41 and over	5	0	1

[1] One unit of product (test sample) at least 1½ yards long and the full width of the cloth shall be taken from each sample roll.

4.1.2 *Preproduction sample inspection.*—When a preproduction sample is required, it shall be inspected and tested as specified in 4.2.2.

4.1.3 *Yard-by-yard inspection.*—Each of the samples selected in accordance with table II shall be visually inspected and defects classified in accordance with 4.2.3.1.1 except sample size and maximum number of defects acceptable shall be as specified in table II.

4.1.4 *Over-all inspection and inspection for length.*—Over-all inspection and inspection for length shall be as specified in 4.2.3.1.2 and 4.2.3.1.3 respectively. Sample size and the maximum number of defects acceptable shall be as specified in table IV.

4.1.5 *Tests.*—Samples selected for tests in accordance with table III shall be tested in accordance with the applicable methods of Federal Specification CCC–T–191 as listed in table V of this specification. The physical values specified in section 3 of this specification apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test methods. The unit of product shall be 1½ continuous yards full width of finished cloth.

4.1.6 *Compliance criteria.*

4.1.6.1 *Inspection.*—Any piece in the sample for inspection failing to conform to the requirements for which the inspection was made shall be considered defective. If the number of defective pieces in the sample exceeds the acceptance number for that sample, the entire lot from which the sample was taken shall be considered to have failed to meet the requirements of this specification.

4.1.6.2 *Tests.*—Any piece in the sample for tests failing to conform to the requirements for which tests were made shall be considered defective. If the number of defective pieces in the sample exceeds the acceptance number for that sample, the entire lot from which the sample was taken shall be considered to have failed to meet the requirements of this specification.

4.2 Military.

4.2.1 *General.* — The sampling, inspection and testing provisions contained in this section are applicable to inspection by and for the Government. Sampling, inspection, and testing for components or materials described herein shall be in accordance with this specification and subsidiary specifications referenced herein to the extent applicable, except that this specification shall govern in the event of conflict.

4.2.2 *Preproduction sample inspection.* — The preproduction sample submitted in accordance with 3.2 shall be visually inspected for appearance, color and finish. The sample shall be tested for physical and chemical properties in accordance with the methods shown in table V when applicable.

4.2.3 *Inspection for acceptance.*—Inspection shall be in accordance with the provisions of Military Standard MIL–STD–105, except where otherwise indicated herein.

4.2.3.1 *Inspection of the end item.*—Defects found during this examination shall be classi-

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fied in accordance with the list shown in 4.2.3.1.1 through 4.2.3.1.2 inclusive.

4.2.3.1.1 *Yard-by-yard examination.*—The defects listed below shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The unit of product for this examination shall be 1 linear yard. The acceptable quality levels (AQL's) for this examination shall be 2.5 major and 10.0 total (major and minor combined) defects per 100 units (yards). The sample size shall be based upon inspection level II, table III of Military Standard MIL–STD–105. The lot size shall be expressed in units of 1 yard each. The number of pieces from which the sample is to be selected shall be in accordance with table IV. An approximately equal number of yards shall be examined in each piece of the sample.

Glossary No. Fed. Std. No. 4	Defect	Classification	
		Major	Minor
90, 91, 92	Hole, cut, tear or break in body of cloth	X	
78	Smash	X	
11, 33	Broken or missing yarns:		
	Three or more contiguous regardless of length	X	
	Two contiguous, 2 inches or more missing	X	
	Two contiguous, less than 2 inches or single, 2 inches or more missing		X
9, 12, 35	Fine filling bar or light place clearly noticeable	X	
	Open place, crack, or hitchback-opening at widest point three or more filling yarn diameters or sequence of openings resulting in a weak spot	X	
	Clearly noticeable, but opening at widest point less than three filling yarn diameters, or raised places resulting from slack warp ends		X
13, 34	Floats or skips-multiple, ½ inch or more in combined warp and filling directions, or harness skips	X	
	Multiple, less than ½ inch in combined warp and filling directions or single floating over 2 inches or more		X
1, 2, 4, 10, 26	Heavy filling bar, heavy place, coarse or additional yarns, clearly noticeable		X
3, 5, 27, 77	Slough-off, jerked-in filling, slugs, slubs, gouts in either direction, three or more times the thickness of normal filling yarns	X	
	Clearly noticeable but not more than three times the thickness of the normal filling yarn		X
6, 28	Kinks-single, more than 1 inch in length or multiple 1 inch or less in length		X
7, 29	Knot, oversize, or untrimmed ends extending from surface		X
40, 41	Wrong draw, reed mark—clearly noticeable		X
42	Temple mark—resulting in clearly visible marks within minimum specified width of cloth		X
49, 50	Selvage tight—resulting in pronounced slackness or waviness in cloth		X
55, 56	Selvage broken, cut or torn	X	
51	Selvage scalloped—indentation exceeding ½ inch		X
93	Baggy, ridgy, or wavy cloth clearly noticeable	X	

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Glossary No. Fed. Std. No. 4	Defect	Classification	
		Major	Minor
89	Tender or weak place—any	X	
79	Abrasion mark—clearly noticeable fuzziness but not resulting in a weak spot		X
69, 70, 74, 75, 76	Spot, stain or streak: clearly noticeable and 3 inches or more in combined directions	X	
	Clearly noticeable and less than 3 inches in combined directions		X
14, 57, 59, 61	Mixed filling, finishing mark, embedded wrinkle or crease —clearly noticeable	X	

4.2.3.1.2 *Over-all examination.*—Each defect shall be counted not more than once in each piece examined. The unit or product for this examination shall be one piece. The sample size (number of pieces selected as sample) for this examination and the acceptance number (number of defects acceptable) shall be as shown in table IV.

Defects

Selvage curled, folded, rolled, or slack, continuously or intermittently throughout the piece.

Selvage beaded or corded continuously or intermittently throughout the piece.

Overall uncleanness, objectionable odor.

Finish not equal to standard sample.

Width less than specified. (See method 5020, Federal Specification CCC–T–191).

Uneven weaving throughout piece, clearly noticeable.

TABLE IV.

Lot size in yards	Sample sizes in pieces	Acceptance number
Up to 1300[1]	3	0
1301 to 3200	5	0
3201 to 8000	7	0
8001 to 22,000	10	0
22,001 to 110,000	15	1
110,001 to 550,000	25	1
550,001 and up	50	2

[1] If a lot contains fewer than three pieces, each piece in the lot shall be examined.

4.2.3.1.3 *Examination for length.*

4.2.3.1.3.1 *Individual pieces.* — The piece shall be examined for gross length. Any gross length found to be less than the specified minimum length, or any gross length found to be more than 2 yards below the gross length marked on the piece ticket, shall be considered a defect with respect to length. The unit of product for this examination shall be one piece. The number of pieces selected as the sample (sample size) for this examination and the acceptance number shall be in accordance with table IV.

4.2.3.1.3.2 *Total yardage in sample.*—The lot shall be unacceptable if the total of the actual gross lengths of the pieces in the sample is less than the total of the gross lengths marked on piece ticket.

4.2.4 *Testing of the end item.*—The methods of testing specified in Federal Specification CCC–T–191, wherever applicable, and as listed in table V shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test method. The unit of product shall be 1½ continuous yards, full width of the finished fabric. For each characteristic, the acceptable quality level shall be 4.0 test failures per 100 units, and the inspection level shall be level L–1, table VII, of Appendix to Military Standard MIL–STD–105.

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TABLE V.—*Test methods*

Characteristics	Req. ref.	Test method	No. of determinations per individual unit of product	Results reported for each unit of product
Weight, oz./sq. yd.	table I	5041	5	Average of 5 determinations to nearest 0.1 oz.
Yarns per inch	table I	5050	5	Average of 5 determinations to nearest whole number
Breaking strength	table I	5100	5	Average of 5 determinations to nearest 1.0 lb.
Mercerization	3.6	Microscopic	1	Pass or fail
Nonfibrous materials [1]	3.6	2611	2	Average of 2 determinations to nearest 0.1 percent
Shrinkage in laundering	3.7	5550	3	Average of 3 determinations to nearest whole number
Weave	3.5.2	Visual	1	Pass or fail
Yarn ply	table I	Visual	1	Pass or fail

[1] The total nonfibrous material shall be determined by method 2611 utilizing the following formula:

$$\text{Nonfibrous materials content, percent} = \frac{0 - S \times 100}{0}$$

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 *Levels A, B, and C.*—The cotton duck shall be packaged in accordance with applicable requirements for class 3 fabrics of Federal Specification PPP–P–51 (see 6.2.1).

5.2 Packing.

5.2.1 *Levels A, B, and C.*—The cotton duck shall be packed in accordance with the applicable requirements of Federal Specification PPP–P–51 (see 6.2.1).

5.3 Marking.

5.3.1 *Civil agencies.*—In addition to any special marking specified in the contract or order, each shipping container shall be marked in accordance with Federal Standard No. 123.

5.3.2 *Military agencies.*—Shipments shall be marked in accordance with the applicable requirements of Federal Specification PPP–P–51.

5.4 Nonmilitary procurement. — Federal Standard No. 102 shall be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6. NOTES

6.1 Intended use. — Types I, II, and III bleached cotton ducks covered by this specification are intended for use in white uniforms for personnel of the Department of Defense. Type IV bleached cotton duck is intended for use in dress uniform trousers of U. S. Military Academy Cadets.

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6.2 **Ordering data.** — Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Type and nomenclature of cloth required (see 1.2 and 6.4).

(c) Width and length required (see 3.5.1 and 3.8).

(d) Availability of standard sample (see 3.1).

(e) Whether preproduction sample is required (see 3.2 and 4.2.2).

(f) Both the level of packaging and level of packing required (see 5.1, 5.2 and 6.2.1).

(g) Whether marking shall be for civil or military agencies (see 5.3).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices, and alternatives, as indicated in Federal Specification PPP–P–51, may be exercised in the preparation for delivery of this cloth.

6.3 **Standard sample.**—For access to standard sample address the procuring office issuing the invitation bids.

6.4 **Nomenclature.**—The nomenclature of the cloths produced under this specification should follow the listings as shown in the respective stock catalogues of the Departments of the Army, the Navy, and the Air Force (see 6.2).

6.5 This specification supersedes the requirements of the following specifications:

CCC–D–730, dated July 30, 1943.
MIL–D–1645, dated 18 November 1949.

6.5.1 *Supersession data:*

Type I — supersedes class A of Federal Specification CCC–D–730.

Type II — supersedes class B of Federal Specification CCC–D–730.

Type III — supersedes class C of Federal Specification CCC–D–730.

Type IV — supersedes Military Specification MIL–D–1645.

6.6 **Transportation description.**—The transportation description applicable to this item is:

Cotton piece goods, in the original piece.
Carload minimum weight, any quantity.
Motor volume minimum weight, any quantity.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
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Navy—S MC
Air Force.

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